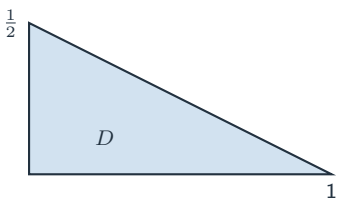
Условие

Нека $f_{X,Y}(x,y) = cx + y$ при $x, y \geq 0$, $x + 2y \leq 1$. Намерете c, $\mathbb{E}[Y|X = \frac{1}{2}]$ и плътността на $Z = X + 2Y$.



Константата c

При фиксирано $x \in (0,1)$: $0 < y < \frac{1-x}{2}$. Вътрешният интеграл:

$$\int_0^{\frac{1-x}{2}} (cx + y)\, dy = cx \cdot \frac{1-x}{2} + \frac{(1-x)^2}{8}.$$

$$1 = \int_0^1 \Big(\frac{c}{2}x(1-x) + \frac{(1-x)^2}{8}\Big)dx = \frac{c}{2} \cdot \frac{1}{6} + \frac{1}{8} \cdot \frac{1}{3} = \frac{c}{12} + \frac{1}{24}.$$

$$\frac{c}{12} = \frac{23}{24} \quad \Longrightarrow \quad c = \frac{23}{2}.$$

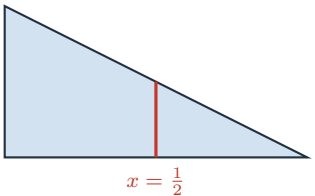
Условната плътност при $X = \frac{1}{2}$

Маргиналът на X (вътрешният интеграл с $c = \frac{23}{2}$):

$$f_X(x) = \frac{23}{4}x(1-x) + \frac{(1-x)^2}{8} = \frac{(1-x)(45x+1)}{8}, \quad 0 < x < 1.$$

Условна плътност: $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$, $0 < y < \frac{1-x}{2}$. При $x = \frac{1}{2}$: $y \in (0, \frac{1}{4})$ и

$$f_X(\tfrac{1}{2}) = \frac{\frac{1}{2} \cdot \frac{47}{2}}{8} = \frac{47}{32}, \qquad f_{Y|X}(y \mid \tfrac{1}{2}) = \frac{32}{47}\Big(\frac{23}{4} + y\Big).$$



разрез: $y \in (0, \frac{1}{4})$

$\mathbb{E}[Y \mid X = \frac{1}{2}]$

$$\mathbb{E}[Y \mid X = \tfrac{1}{2}] = \int_0^{1/4} y \cdot \frac{32}{47}\Big(\frac{23}{4} + y\Big)dy = \frac{32}{47}\Big(\frac{23}{4} \cdot \frac{1}{32} + \frac{1}{192}\Big).$$

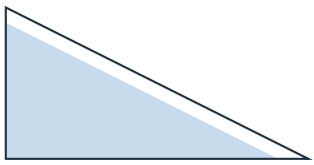
(Тук $\int_0^{1/4} y\,dy = \frac{1}{32}$ и $\int_0^{1/4} y^2 dy = \frac{1}{192}$.)

$$\mathbb{E}[Y \mid X = \tfrac{1}{2}] = \frac{32}{47} \cdot \frac{71}{384} = \frac{71}{564} \approx 0{,}126.$$

Условната плътност е почти постоянна (доминира $\frac{23}{4}$), затова очакването е малко над средата на интервала, $\frac{1}{8} = 0{,}125$.

Плътността на $Z = X + 2Y$

За $z \in (0,1)$: $\{Z \le z\}$ е по-малкият триъгълник $x + 2y \le z$:



вложени триъгълници $x + 2y \le z$

$$F_Z(z) = \int_0^z \int_0^{\frac{z-x}{2}} \Big(\frac{23}{2}x + y\Big)dy\,dx = \int_0^z \Big(\frac{23}{4}x(z-x) + \frac{(z-x)^2}{8}\Big)dx$$

$$= \frac{23}{24}z^3 + \frac{1}{24}z^3 = z^3.$$

$$f_Z(z) = 3z^2 \mathbb{1}_{(0,1)}(z).$$

Отговор. $c = \frac{23}{2}$; $\mathbb{E}[Y|X = \frac{1}{2}] \approx 0{,}126$; $f_Z(z) = 3z^2 \mathbb{1}_{(0,1)}(z)$.